

ROBIN INHO KEE

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🏠 [Website](#)

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Education

University of Michigan

Doctor of Philosophy in Robotics

Aug 2025 – Present

Ann Arbor, MI, USA

- **Advisors:** Prof. Vasileios Tzoumas and Prof. Necmiye Ozay
- *Draper Scholars Program* – fully funded Ph.D. fellowship in autonomy (**Draper advisor:** Dr. Begum Cannataro)

University of Michigan

Master of Science in Mechanical Engineering

Aug 2023 – Dec 2024

Ann Arbor, MI, USA

Yonsei University

Bachelor of Science in Mechanical Engineering

Mar 2016 – Feb 2022

Seoul, South Korea

- *Mandatory military service: Republic of Korea Air Force, Oct 2020 – Jul 2022*

Research Interests

Multi-Robot Autonomy, Communication-Aware Coordination, Learning-Enabled Control, Scalable Decision-Making

Publications

* *Equal contribution*

Journals

- [J2] Disentangled Prototypical Convolutional Network for Few-Shot Learning in In-Vehicle Noise Classification.
R. I. Kee, D. Nam, S.-J. Buu, and S.-B. Cho. *IEEE Access*, 2024. [Paper]
- [J1] “Measurement of ankle joint movements using IMUs during running.”
B. Kim, S. Hong, I. Oh, Y. Lee, **I. Kee**, and S. Lee. *Sensors*, 2021. [Paper]

Conferences

- [C5] “Time Shift Governor-Guided MPC with Collision Cone CBFs for Safe Adaptive Cruise Control in Dynamic Environments.”
R. I. Kee*, T. Kim*, A. Girard, and I. Kolmanovsky. *IEEE CCTA*, 2025. [Paper]
- [C4] “Learning to Refine Input Constrained Control Barrier Functions via Uncertainty-Aware Online Parameter Adaptation.”
T. Kim, **R. I. Kee**, and D. Panagou. *IEEE ICRA*, 2025. [Paper]
- [C3] “Constrained Control for Autonomous Spacecraft Rendezvous: Learning-Based Time Shift Governor.”
T. Kim*, **R. I. Kee***, I. Kolmanovsky, and A. Girard. *AIAA SciTech Forum*, 2025. [Paper]
- [C2] “Dynamic Prototype-guided Memory Replay for In-Vehicle Noise Classification.”
D. Nam, **I. Kee**, S.-J. Bu, and S.-B. Cho. *Korea Data Mining Society*, 2023. **SAS Student Paper Award**. [Project Page]
- [C1] “An Intelligent Lock-Out Tag-Out System for Monitoring and Control of the Locked Device.”
W. Jo, S. Hwang, **I. Kee**, I. Lee, and S. Lee. *IEEE IEEM*, 2019. [Project Page]

Manuscripts Under Review / Preprints

- [P4] “AC-DC: Adaptive Communication for Scalable Dynamic Average Consensus in Multi-Robot Ergodic Search.”
R. I. Kee, B. Cannataro, and V. Tzoumas. *IEEE ICRA*, 2027 (submitted). [arXiv]
- [P3] “Learning to Adapt Control Barrier Functions Under Epistemic and Aleatoric Uncertainty.”
T. Kim, **R. I. Kee**, and D. Panagou. *Preprint / under review*. [arXiv]
- [P2] “Toward Daily-Life Sarcopenia Identification via Real-Time Gait Assessment Using a Wearable Hip Assistive Robot.”
S. Kim*, S. Kim*, J. Kang, **R. I. Kee**, K. R. Mun, Y. Lee, K. Kim, Y. Cha, and J. Lee. *IEEE RA-L* (under review). [Project Page]
- [P1] “Vision-Ultrasound Robotic System based on Deep Learning for Gas and Arc Hazard Detection in Manufacturing.”
J.-H. Lee, D. Nam, **R. I. Kee**, Y. Kim, and S.-J. Buu. *arXiv preprint*, 2025. [arXiv]

Research Experience

Graduate Research Assistant, University of Michigan

Aug 2025 – Present

Intelligent Robotics and Autonomy Lab (PI: Prof. Vasileios Tzoumas)

Ann Arbor, MI, USA

- Conceived, designed, and implemented AC-DC, a peer-to-peer dynamic consensus framework that jointly adapts communication partners, timing, and consensus-state blocks under range, rate, and interference constraints
- Evaluated AC-DC across 12 settings with 20 paired trials each; reduced mean paired AUC by 29.5% and 24.5% versus ADMM-DAC and PP-ACDC, while those baselines used $5.8\times$ and $11.5\times$ as much modeled payload [P4]

Graduate Research Assistant, University of Michigan

May 2024 – Aug 2025

Distributed Autonomous Systems and Control Lab (PI: Prof. Dimitra Panagou)

Ann Arbor, MI, USA

- Implemented and integrated the OA-CBF research pipeline by adapting control and learning components and extending the codebase with graph-attention and uncertainty-aware screening for online CBF-parameter selection
- Conducted the complete evaluation across five nonlinear systems; achieved 0% safety failures and 100% goal-reaching in three 100-trial benchmarks, with 39–65% shorter reach times than the safe fixed-parameter baseline [C4/P3]

Graduate Research Assistant, University of Michigan

Jan 2024 – Feb 2025

VODCA Lab (PIs: Profs. Anouck Girard and Ilya Kolmanovsky)

Ann Arbor, MI, USA

- Developed the LSTM prediction module for a hybrid Time Shift Governor with online verification and a bisection fallback; reduced average time-shift computation by 84% in LEO and 70% in Molniya-orbit spacecraft-rendezvous simulations [C3]
- Designed and implemented a TSG-guided MPC-CBF controller for adaptive cruise control in dynamic curved-road scenarios; achieved 100% success over 50 randomized trials versus 82% for baseline MPC-CBF [C5]

Research Intern, Korea Institute of Science and Technology

Jul 2022 – Jul 2023

Assistive and Interactive Robotics Lab (PI: Dr. Jongwon Lee)

Seoul, South Korea

- Built the initial end-to-end sensing and learning pipeline for a 4-DOF wearable hip robot, including multimodal sensor fusion, motion-capture synchronization, preprocessing, and foot-trajectory and gait-parameter estimation
- Established the experimental pipeline and led initial human-subject studies; the resulting 56-participant, 51,610-stride study achieved 6.69 cm stride-length RMSE and 90.17% stride-wise sarcopenia-identification accuracy [P2]

Technical Skills

Programming & Tools: Python, C/C++, MATLAB, PyTorch, Linux, Git

Robotics & Methods: ROS/ROS 2, MPC, CBFs, Online Learning, Multi-Robot Coordination

Selected Awards and Honors

Draper Scholars Program (Full Ph.D. Fellowship), Charles Stark Draper Laboratory, 2025–Present

SAS Student Paper Award, Korea Data Mining Society, 2023

1st Place, International S.M.A.R.T Startup Competition, Innovative Technology and Energy Center, 2020

Science and Engineering Undergraduate Internship Program Scholarship, Yonsei University, 2020

Academic Highest Honors, Yonsei University, 2019, 2020

2nd Prize, Robot Open Source Lab, Samsung Open Source Conference, 2019

Academic Excellence Scholarship, Yonsei University, 2018, 2019

Teaching & Academic Service

Graduate Student Instructor: AEROSP 740, Online Learning for Single- and Multi-Agent Control

Fall 2026

- Developed lab materials and code for quadrotor simulation and Crazyflie hardware experiments; held weekly office hours and supported simulation and hardware assignments

Reviewer: CCTA 2026 **Conference Volunteer:** L4DC 2025

Leadership

Roboin Robotics Club

Jun 2018 – Aug 2023

President (2019); Advisory Committee (2020–2023)

Yonsei University, Seoul, South Korea

- Selected projects: Autonomous Fire-Extinguishing Robot [Project Page]; Theo-Jansen Legged Robot [Project Page]